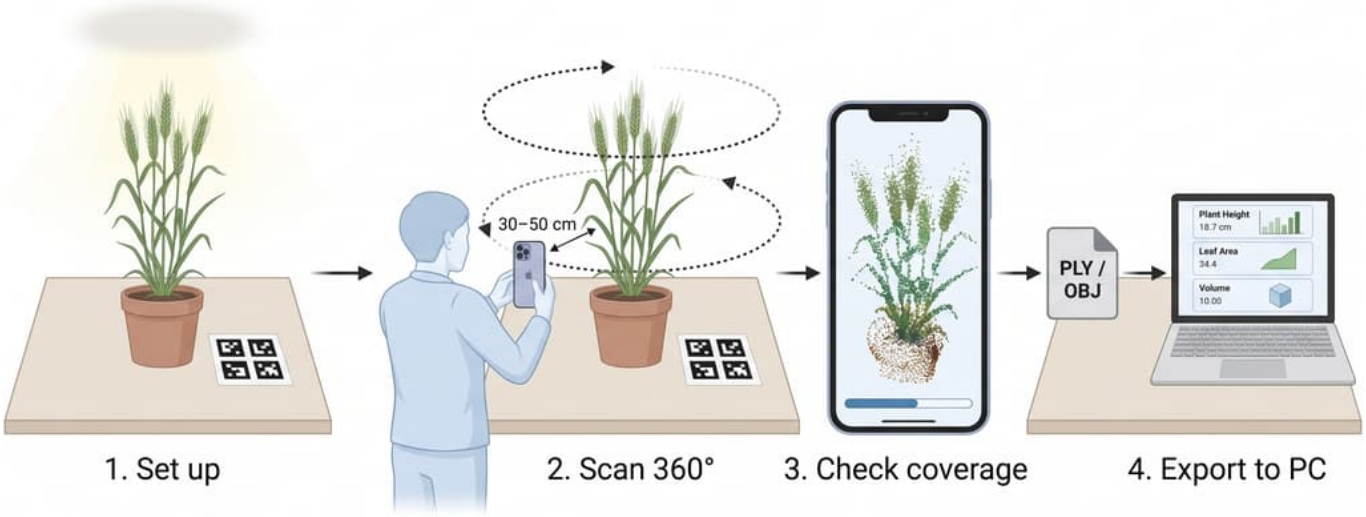
SOP-7 iPhonotype app (iPhone)

Guided orbit capture on the phone, analysed in PhenoForge from the .ftscan package

SOP-7 · PhenoForge capture SOP · version 1.0 (October 2026)



iPhonotype guides you around the plant and takes 90–140 photos. On the phone it then builds a textured mesh (Apple photogrammetry, USDZ), a 3D Gaussian-splat cloud (PLY, in metres) and preliminary traits, and exports everything as one **.ftscan** package (a zip file). PhenoForge opens the package directly and measures the splat cloud. A test scan took 2.6 min to capture and about 4 min to process on the phone. LiDAR is not needed: the app uses the rear camera and the motion sensors.

What you need

Checklist

- iPhone with the iPhonotype app (tested with app version 2.3.1, iOS 26)
- ArUco card (DICT_4X4_50, 50 mm) beside the pot; the app also recommends markers for scale checks
- Plain, matte, non-green background and diffuse light (the test scan had about 600 lux)
- About 50 cm of free space all round the plant, and no wind
- A way to copy files to the PC (AirDrop to a Mac, OneDrive or USB): a package is about 250–300 MB

Capture

1. **Set up**
   Put the pot on a matte surface with the ArUco card flat beside it, in view from every side. Put the plant label where the camera sees it.

2. **Start a scan and label it**
   Start a new **Spatial Orbit** scan. Fill in the plant ID, genotype, treatment and operator: they are stored in the package (manifest.json) and are otherwise empty.

3. **Orbit**
   Follow the app's guide: one slow, close orbit, then a **higher pass** and a **lower pass**. Keep the whole plant and the card in the frame. Aim for about 140 photos; 90 is the app's minimum.

4. **Let it process**
   Wait until the mesh and the splat are finished on the phone before you export. Keep the phone plugged in and out of the sun so it does not slow down from heat.

5. **Export**
   Export the **.ftscan** package. Either copy it to the PC (AirDrop, OneDrive or USB), or send it over Wi-Fi: in PhenoForge choose **File > Receive from phone (Wi-Fi)**, press **Start**, scan the QR code with the iPhone camera and pick the package on the web page that opens. You do not need to unzip it.

In PhenoForge

1. **Open the package**
   On the **Start** tab click **Analyse an iPhonotype scan** and choose the .ftscan.zip (or drop it on the window). PhenoForge stands the splat cloud upright, takes the metric scale, plant ID, genotype, treatment and light readings from the package, and runs the analysis straight away. The **vs iPhonotype** tab compares its traits with the app's; the **Package** tab lists what the app recorded.

2. **Clean and frame**
   Leave *Input is Y-up* **unticked**: the package is already turned upright on loading, and ticking it lays the plant on its side. If you open the USDZ mesh on its own, also leave it unticked. Keep *Remove outliers*, *Fit ground plane* and *Remove ground points* ticked, and use vegetation index *exg* to remove the pot and soil.

3. **Check the scale**
   The package scale is used automatically (*Already metric*). With a card in view, switch to **ArUco markers**, type the measured marker size and press **Run analysis** again; **Markers image** shows each marker's size. Without a card, *Two picked points* across a known distance such as the pot rim also works.

4. **Check the model**
   Check vs photos draws the 3D scan over each of the app's photos and reports the overlap (IoU). Most frames above about 0.6 means the model matches what the camera saw.

5. **Save**
   Save results writes JSON and CSV files, including the comparison with the app and the package details. For many scans use **File > Analyse a whole folder**: one row per scan in all_scans.csv.

What is in the package

File	What it is	Use in PhenoForge
splats/iphone_splat_reconstruction.ply	Gaussian-splat cloud in metres, scaled from the camera path (confidence in 3dgs/workflow.json)	Default. Its height matched the app's plant height within 1.5 cm in the test scan
mesh/photogrammetry_mesh.usdz	Textured mesh in Apple reconstruction units; includes the pot	Optional. Height includes the pot unless it is filtered out
analysis/on_device_*.json	The app's preliminary traits and top-view plant masks	For comparison only; the package says desktop results replace them
rgb/, 3dgs/images, 3dgs/transforms.json	Photos and camera poses	Re-run in COLMAP or Nerfstudio if needed

Quality checks

Check	Pass	If it fails
Photos in the package (logs/native_object_capture.json, image_count)	≥ 90, ideally 140	Rescan with a slower orbit
Coverage (manifest.json, quality.coverage_estimate)	≥ 0.8 (suggested)	Add the higher and lower passes; avoid gaps behind leaves
Plant stands upright after the ground fit	upright	Untick Y-up. If the splat .ply was opened on its own (not the package), tick Y-up. Crop away a tilted table or the background
Marker size after scaling	within ±2 %	Use ArUco scaling; check that the card is flat
Plant height vs ruler	within 1 cm	Check the scale; fit the ground plane on soil, not on leaves

Good to know

- The app's on-device traits are a preliminary estimate. Use the PhenoForge results, and validate them against manual measurements (Validation tab) before relying on them.
- The app may warn that metric scale is not verified: that refers to the mesh. The splat cloud carries the camera-based scale. Still put a card in view, because it is the only independent check.
- Projected area and leaf area differ between the splat and the mesh because sparse leaf edges are treated differently. Compare a trait only between scans analysed the same way.
- The app reads any text in view as a possible tray label (in the test scan it read safety signs). Type the plant ID rather than relying on it.
- Leaves and awns thinner than about 3 mm can be thickened or lost.